GM107: What If AI Actually Pushes Interest Rates Higher? ft. Matt Klein

Podcast: Top Traders Unplugged

AI Executive Summary

Core Thesis

The episode explores the current US economic landscape through the lens of inflation, productivity, and debt sustainability, arguing that while conditions are different from the late 1990s, risks remain elevated due to capital-intensive AI spending and global imbalances. Matt Klein posits that despite the Fed's recent constructive pivot (Kevin Warsh), real interest rates may rise or remain high to reflect capital costs rather than support deflationary policy. Furthermore, China's persistent structural trade surplus is a destabilizing force not easily addressed by tariffs alone, necessitating long-term vigilance regarding currency interventions and global capital flows rather than assuming a repeat of the 1990s productivity boom.

Key Arguments & Data

- **Fed Policy & Warsh's Tenure:** Kevin Warsh's Jackson Hole speech was constructive regarding inflation targets and the need to close inflation gaps, countering prior concerns about his "hack" reputation or AI deflationist biases. Markets reacted with expectations of short-term rate hikes despite a seemingly stable job market.
- **AI & Productivity Paradox:** There is no meaningful productivity acceleration in current data (unlike 1997). AI investment is capital-intensive, raising input costs for memory chips and consumer goods (e.g., Apple raising MacBook prices by ~20%), which pushes inflationary pressure rather than reducing it.
- **Interest Rate Logic:** Bond yields are justified not by deficits or debt service alone, but by the gap between high growth expectations and real returns investors demand. The 1990s experience showed investment booms correlate with rising real interest rates because higher productivity attracts more capital demand.
- **Yield Levels:** The 10-year yield has trended upward since 2023, reaching levels comparable to 2007 or late 2023 but below early 90s highs. Real rates remain lower than 1990s peak levels despite higher

debt-to-GDP ratios historically because of low interest rate lock-ins from the 80s.

- **US Debt Sustainability:** Unlike Brazil's high real-rate environment (hyperinflation legacy), the US situation is stable because G (growth) > R (interest rate cost). However, current account deficits mean foreigners finance this by buying US assets, keeping yields relatively manageable compared to default risk in smaller economies.
- **China & Global Imbalances:** China's domestic consumption is structurally low relative to production; trade surplus has exceeded $1 trillion since pre-pandemic levels. This offset requires export demand from the rest of the world and creates friction for trading partners (e.g., EU).
- **Trade Policy Realities:** Tariffs on China may not be effective due to rerouting effects (imports shifting to Southeast Asia) similar to 2001-2002 era. The Plaza Accord is politically toxic for Chinese leaders despite it being a useful quantitative calibration tool.
- **Currency Interventions:** The Treasury intervened in Japan's yen market (~$500m reported), likely out of fear of raising US Treasuries via Japanese sales. Japan seeks stronger Yen to mitigate import costs but faces domestic inflation/wage stagnation constraints on interest rate hikes.
- **Fed Task Forces:** New task forces (including Balance Sheet, Productivity View) address internal Fed ambiguity. The Balance Sheet task force features experts like Raghuram Rajan and is particularly interesting due to unsettled policy questions regarding the Fed's balance sheet reduction path.

Actionable Takeaways

- **Investment Horizon:** Expect elevated real interest rates to persist as capital-intensive AI investment outpaces productivity data; do not expect a deflationary spiral from current tech spending.
- **Fed Watch:** Monitor the Balance Sheet task force (Rajan, Stein, etc.) outputs closely for potential policy shifts regarding balance sheet reduction strategies and liquidity management.
- **Risk Management:** Be wary of "Brazil-like" equilibria where real rates become excessively high relative to growth; while not currently

present, it remains a structural risk in emerging markets with legacy hyperinflation histories.

- **Geopolitical Outlook:** Do not view tariffs on China as a sustainable solution given the rerouting capabilities of global supply chains; focus on how currency intervention and capital flows impact exchange rates rather than direct trade volume changes.
- **Yield Strategy:** Consider bond yields to be justified by strong growth expectations rather than solely inflation concerns, but remain cautious about potential future volatility in both sectors as productivity data remains speculative.

Full Transcript

So it's a really bad equilibrium that they find themselves stuck in. And that's a legacy of having several hyperinflation episodes over the past few decades. They haven't had any since then, but I think that's a big explanation for that. We're not there. I hopefully won't be there. I mean, that is a potential situation to be wary of. But I don't think we're anywhere near that right now. Welcome to Top Traders Unplugged. In markets, success doesn't come from predicting what happens next. It comes from being prepared for what you can't predict. In each episode, we go deep with some of the world's most thoughtful minds in investing, economics, and beyond to understand how they think, how they prepare, and how they decide, and the experiences that shaped how they see the world. No noise, no shortcuts, just real conversations to help you think better and invest with confidence.

Welcome back to Top Traders Unplugged. My name is Alan Dunne and today I'm delighted to be joined by Matt Klein. Matt is the author of the Overshoot research publication. He's previously been a journalist at the Economist Bloomberg Barrens and at FT Alpha Bill. He's the author with Matt Michael Pettis of Trade Wars or Class Wars. And he's a previous guest here from Top Traders Unplugged. Matt, great to see you. How are you doing? Very good. Thank you for having me, Alan. Great to have you back and as I say you were a previous guest from I think it was started last year so if people want to go back and hear about your backstory they can check it out there but obviously things have evolved a lot since the last 18 months I guess and I guess most notably with the Fed in particular and as we're just recording a week after Jackson Hole maybe that's a good place to jump off from.

I mean, what were your impressions of Kevin Warsh at Jackson Hole? I thought the speech itself was very reasonable. I think it's also important context of what, you know, I think when he went in, there were a lot of understandable concerns about, you know, what the nomination process was, how he got the job, or the criteria that were used to pick him in the first place. And the concern, I think, not unreasonably that, given the criticism that had been levied at the Fed by President Trump and by a lot of

other people in this administration that they were going for someone who essentially would be a hack who would just do whatever they want and get interest rates down. And initially he didn't really do a lot to assuage those concerns. I think the announcement in particular of the productivity and jobs task force was the one that kind of stuck out while a lot of the task forces that had been announced a few months ago.

all seem like very reasonable subjects with very credible people. That one kind of had some red flags of what they're maybe trying to get a foregone conclusion there. Before being nominated, he'd been saying that the advent of AI related technologies would inherently be deflationary and create the scope for lower interest rates. So there's a concern again, trying to sort of predetermine the answer. I think it was very encouraging that a Jackson Hole, unlike at his first Fred Press conference, he, I think, did a lot to kind of address some of those concerns. First of all, being very clear about what the inflation target actually was. You know, his first press conference, he was a little vague about what that, you know, whether the inflation target actually was what they had said before, whether it be to something different. Also very clear that, you know, they would need to do something to get the inflation, actual inflation to get to where the target is, that there is a gap there. He was, I think, it was encouraging that I think His analysis of the sort of current economic situation is actually very similar to what I've been saying for a while, which is that, look, the job market seems pretty good on the whole, pretty stable. Businesses are investing, consumers are spending, and inflation is not obviously decelerating past a certain point. I mean, obviously, if you're comparing it to say the first half of 2022, we've made a lot of progress, but there hasn't really been a lot of progress in the past sort of three years.

A lot of Fed officials had been slow to realize that because they were hoping I think that they're, you know, the rest of whatever had occurred in 21 and 22 were just eventually unwind and it hasn't happened yet. And I think they're now starting to realize maybe there's something more fundamental and they will need to adjust policy. So I think that's one of the big takeaways I think we got from that speech, which I think is constructive. We'll see what happens the next Fed. few Fed meetings, but the market pricing reaction was pretty clear. The assumption, okay, maybe

the Fed is going to raise it short-term rates a little bit more from here. You mentioned his whole take on, I guess, AI and the productivity boom, et cetera. And I know that's something you've been writing a lot about and people have drawn parallels between now and the 90s and whether they're valid or not.

And then you were involved with Sebastian Malaby as he wrote his book about Greenspan, so you've read all of those transcripts. So maybe give us your take on what comparisons are relevant with the 1990s or not, and what does all that mean for policy? Sure. Well, I guess the first point to bear in mind is that so far we've not actually seen any meaningful productivity acceleration in the data so far. That doesn't necessarily mean there won't be one. If you go back to the 1990s example, it's instructive that even as late as, say, 1995, when, you know, I mean, I was young then, but I was alive. I mean, the computer and the innovations of the computer internet were very much visible. But the productivity data didn't really start accelerating until 1997.

Greenspan has his reputation of having been sort of brilliantly anticipating a productivity boom. He didn't start making that argument until the middle of 1996. So he was in front of the official data, but not radically in front of it and certainly not compared to where we are now. Now that had been said, productivity might very well accelerate and that's fine. But it's just important context and so far everything's speculative. What has happened now, what we do know, as of now is that there has been a massive increase in business investment. And that has pushed up the prices of certain goods that are used as inputs for data centers dramatically. Most notably memory chips, but not only memory chips. And that has started the filter through to consumer prices. The consumer prices of I think Apple just recently announced they're raising all the prices on their MacBooks and iPads by 20% or something, right? But even before that it was showing up in things like people buying flash drives or memories and input for computers, right? So that's having an impact, direct impact. There's all sorts of indirect impacts as well. I live in San Francisco and you can tell by living here the housing market has definitely had a result of the spillover from the wealth boom of the AI frontier companies.

But okay, leave all that aside whether there's going to be a productivity boom or not. If we just take the rerun the tape and say, we're going to have the 1990s. I think it's very important to know two things. One, the level of interest rates in the 1990s before the productivity boom really began was substantially higher than it is now, even though at that point in time inflation was running slower than it is now. Overall, nominal growth was actually about the same because real growth was stronger, but inflation was slower. And then, once the productivity boom started, as best as we can tell, real industry rates went up. It's a little hard to know for sure because the instruments we have now were not as sophisticated back then. The Treasury and the Pleasure Protective Securities were very new, so it's hard to know for sure. But it seems like yields went up, real rates. And from levels that are already elevated compared to where we are now.

What I would also say is important is it may not have gone up enough for what they should have given what was going on in the economy. You expect that if productivity is accelerating, if there's suddenly so many more attractive investment opportunities as a result of these new technologies, then the demand for investment would rise relative to everything else. And that should make the cost of capital go up because your expected returns are going to go up because productivity is increasing. And that did happen somewhat, but arguably it could have happened more. then. But again, if you're happening now, even if you think inflation is not going to, is going to decelerate somewhat, this sort of real growth impact should be quite substantial. And again, this is the starting points are also different. And then, you know, one last point I'll add, I know I said, I think I'll start with two, so I'm going to add like three, but like inflation didn't actually slow that much as a result of the productivity boom in the 90s. The productivity boom in the 90s mostly manifested as more growth.

rather than less inflation. To the extent there was a slowdown in inflation, it was largely because of things like commodity prices falling a lot because you had these very large crises in emerging market countries and that was very destructive to demand. Japan was kind of going through the worst of its post bubble period. And that dragged down prices and then the dollar was rising, it dragged down imports. None of that was actually sort of productivity related. And so even if you were to get a repeat of that, and

maybe we're talking like a percentage point, on the inflation side for the US, but there's so many countervailing forces now that it'd be very skeptical. I mean, maybe it happens, right? Like there are a lot of reasons to think it's the opposite, right? I mean, since COVID and Russia's war in Ukraine, and now the war in Iran, I think there's really an impulse for countries around the world to be trying to stockpile more physical goods and increase their domestic productive capacity, which should, I would think sort of be...

more inflationary than not. So at least it's the reverse of sort of the impulse we saw in the 90s. And so, you know, again, like I don't think like we're going to necessarily have repeat. I mean, I don't know, right? Like AI is changing very rapidly. I'm not an expert in this. Even people who are experts, I don't think they're, they know enough about the flow through to the macro variables, but even if you were to sort of naively say it's the same thing, the implication that interest rates would go down, I think is, is not correct. I think it's very plausible that if that were to happen, interest rates would go up. I mean, you mentioned there, you know, 1995, people started using more computers or we started, you know, bring those into the workplace more. So, I mean, just like me, there was kind of a couple of different waves in the 90s. Obviously, I had that kind of greater use of computers and investment in computers. And then towards the end of the 90s, obviously the whole internet thing and, you know, investing in.

Internet infrastructure as well. So I suppose that you've got kind of demand side elements to it and supply side elements and they come at different times. I mean, I guess the argument with AI is, you know, if we went back maybe last year, there was a lot of debate, people were skeptical about, okay, is this really going to be a thing? And then as we went through this year, people started to say, oh no, it really is, you know, they started to see the applications and Claude or whatever else and and as much more tangible sense of okay, this is really going to be a meaningful impact on productivity, even if we haven't seen it in the data yet. I mean do you think there is that argument that we could potentially see that productivity impact from this new technology quicker? Obviously like the internet impact came maybe a decade later, whereas in this case people are anticipating it 12 months, 24 months, 36 months time. It's very hard to judge and there are a couple of things In general, the diffusion from the invention of a new

technology to the actual manifestation of that, the productivity statistics usually takes a long time. Because if you, the people who were relatively early adopters of using AI to improve their workflows, my understanding is the most software engineers because it's very good at helping you write code. The question is, well, the two things. One is, has that particular process of improving sort of software coding already hit its limit, or is there like a lot more of that to run? And then the other one is to what extent can other industries have comparable benefits? And it's, you know, how long will it take people to learn how to do that? I mean, the sort of canonical example people have been talking about for years is electricity, right? People discovered how to use electricity electric motors and stuff. It took decades before people understood all the ways they could update their existing processes to be more efficient and fully take advantage of what the advent of electric power and electric motors will let you do. I don't know if it's going to be that slow for AI related technologies, but there are, you know, you can call them bottlenecks, call them limits, whatever, but like as long as humans are making these decisions, which some people might say that will not be the case, but if humans are making decisions, it takes a long time for things to roll out. And you can look at way most, right? Driverless cars, like San Francisco has been at the forefront of allowing these, and it's now very easy. You see them driving around all the time. A lot of other jurisdictions have not allowed them for various reasons. A lot of people don't even know that it's possible to have driverless cars at scale being a practical technology in most of the world.

I don't know if it's going to be like that necessarily, but the diffusion from this technology exists to figure out how to best use it, that's really the challenge and how applicable will it be across industries. As I said, for software coding, it's very clear. I think people figured out a lot of the ways to make it work for them. But is that going to, that's only a small part of the economy, right? The extent to which you can boost economy productivity, that's harder. I guess separately, and this is hopefully not too much of a tangent, Productivity itself is not something we can measure. As a micro-economist, you can look at specific tasks and how long it takes and you can say, that's productivity. But at the economy-wide level, there's no way of doing that. All you do is you say, well, here's the total amount of money spent in the economy or the amount of money earned in the economy.

Here's some inflation measure we use to turn that into real growth or real output. And then we have some estimate of how many hours people are working and you just divide these things and then that's productivity.

But obviously there's lots of room for measurement error across all three. It's not really clear why something will change one way or the other. You know, output per hour can adjust for reasons other than what we might think was real productivity changes if there's like big shifts in the composition of the economy. I mean, the most extreme cases during the pandemic, lots of people working in relatively low paid jobs became unemployed because there were restaurants and retail stores and stuff like that. So the composition of employment of whoever was left working suddenly skewed much higher income. And so doing the sort of naive math of productivity looked as if productivity just exploded. We had this great productivity boom. That is not what happened, right? Maybe there were some changes during the pandemic that did lead to lasting productivity increases and like with remote work or what have you, but like that's not really what the data we're showing at the time. And so I think it's very difficult to say over any kind of short-term horizon what's going on productivity. In the 90s, you could see over time there was this kind of notable shift on the growth rate.

But again, it's decomposing exactly what happened. People have spent decades later. I think now they have a sense of what happened, but at the time it was not easy. And so now, again, we haven't even seen an acceleration in productivity growth yet. But if you were figuring out exactly, it would be challenging. And just on that, we did seem to see stronger productivity growth for a while, but then that's moderated. Is that correct to say? Yeah. I mean, again, part of it is that post-pandemic period has looked really weird because you had this big increase in productivity. in the depths of the pandemic 2020. And then a lot of people get rehired, the economy reopens, all this good stuff is happening. It looks like productivity is falling.

which is a very rare occurrence. But again, this is mostly because people are able to go back to restaurants now. That's a lot of bad thing. And then eventually that process kind of stops and you get back to some sort of

normal trend. And then it looks like productivity is growing a bit faster, but it's really hard to tell like really what's going on. You have to kind of zoom out and look at a trend like, okay, it's not really doing much. Maybe a little bit, like by the end of 2024, you were starting to pull a little bit ahead of the trend. It's hard to say. And then we've kind of been more or less on trend. Another challenge here, so the The Federal Reserve Bank of San Francisco has this method of decomposing productivity further in different parts because basically the idea is if output per hour, which is the normal thing of productivity, as I said, you take like this whole amount of stuff, produce the economy and divide it by hours, work, see what changes, that can be misleading. So they try to decompose into different parts. Say how much of that is due to say the workforce becoming more experienced, for example, right? Like that's good, but that's not productivity. That's just like We're older and wiser or whatever right that that's like a different thing or we have more machines available to help us with stuff That's good, but it's that's just that's not doing more with less. That's doing more with more right? It's a different kind of phenomenon so they kind of try to strip out those elements and then they have this other thing called resource utilization which is the idea is that During periods when kind of the economy is running hotter it might look like growth is gutting a little bit faster because people are just working harder But it's not showing up in other forms of you know worked in and they kind of try to strip that out and you can argue about how they do it But they have those methods. So what's ever is left is sort of underlying measure productivity and their estimate is that actually there's been zero growth in underlying productivity in the past like 18 months, maybe it's actually down slightly um or more than 18 months, right? So basically no growth there. All the growth we've seen is in these other factors. And so, I mean, we can quibble with the methodology and the data or whatever, but that suggests that in fact, that that's actually unusually bad. And maybe over the short time horizon, it's not enough to be meaningful, but I think it's instructive. What their model says and what they've written about is that so far, it seems that people are working harder rather than smarter so far, which is kind of the opposite of what you'd think naively based on.

we're seeing with the technology. You mentioned the task force and obviously Warsh does seem to be a believer in AI and based on what he said before he came into the office or into the seat. And he did mention in his

speech last week, he mentioned the idea of AI being maybe an additional factor of production or a separate fact. If you were to treat it as a separate factor of production, does that change anything or does that change economic models in any way or is that just a different label? I don't think so. I mean, there already are methods for incorporating intermediate inputs, essentially. You could think of tokens as being energy or some other kind of material input.

I mean, of course tokens are produced with energy. So that's like, you know, maybe it's double counting a little bit. So like, I don't know. I mean, like, I know, I do know, because I've talked to them, the economists at the, the BEA, which is the agency that produces the GDP statistics and things like that. They're aware of these issues they're thinking about. They've published papers about how to look at AI, you know, how that creates, you know, how it translates to value, like if a company is buying an AI product. or AI services and use those services to produce a thing, like how that values distributed economies. They're aware of these things. I don't think it's something that's just completely missing. I think it's just so far it's not really been much of an issue yet. I mean, the key thing is, the reason it's not being missed, which is very important, is that this costs money. If people were just giving this stuff away, maybe it would be harder to judge, but if companies are paying for things, the government is very good. The statistical agency is globally very familiar with if companies buy things, How that shows up in the national accounts, you know, you're buying and it's cost of goods sold essentially And you can trace that and someone's getting profits and someone's having an expense and someone's getting income and they know how to do that So I don't think it's fundamentally new in that sense in general I'd be very skeptical because I've seen this before of people saying this like new technologies Changes our fundamental understanding how the economy works The fundamental understanding of how the economy works is pretty robust to new technologies showing up. You just have to figure out how to classify them, but I think so far we've had a pretty good track record. Yeah. I mean, just in relation to the ongoing spend on AI, the data center build out and all the associated activity, obviously the sense in the market is that has just been revised higher and higher and it's having a more meaningful impact on GDP. Now, obviously there is this concern, okay, is this a kind of going to

be some kind of boom bus scenario? Obviously the dot-com was different. It was more valuation led here. We're having real investment. I mean, in the sense that the dot-com might have been a price bubble. Now we're talking about a possible earnings bubble in the market driven by strong capex. I mean, what's your perspective on that? Do you see this as being a durable investment trend or are you concerned that will reach a point where A lot of these data centers are built out and suddenly see a dramatic slowdown in that kind of spend at some point. There's a lot of questions in there. I mean, the first thing I would say is that I mean, the 90s, I think it's useful to go back and look at these things because there actually was a real investment boom there. I think it's very easy to say it was a dot com phenomenon. It mostly showed up in like the valuation of stocks like pets.com or what have you, but you can strip that back.

And there was, if you look at things like investment net of depreciation and inflation, real net investment, you had the biggest boom in the 90s basically over any sustained period since the data began in 1929. Maybe like a little bit during World War II, maybe there was like a little lead up to it, but like it was massive. The Federal Reserve has a series on industrial capacity. That goes back to the 40s. The growth rate in that period was again basically unprecedented. And that's even if you exclude investment in what they call high tech sectors, so like semis, computers, things like that. So it was actually a very broad based boom in CapEx. And that was also separate from things like the laying of fiber optic cable at the end of that. So there really was a real investment. I think part of the reason why the bust coming out of the 90s was as destructive as it was. I think more than a lot of people appreciated at the time even now is because you had this big collapse in business investment. The government, the Fed tried to offset that through essentially a housing bubble, which I don't think was really sustainable.

But the fact that they had to do that and you still had relatively anemic growth in the 2000s is I think demonstrates just how big that was a problem that was. So I do think it's a real concern for now. The main sort of positive I would say this time, so in the 90s, all that investment mostly translated to additional domestic production. Most of the stuff that companies were buying was being made in the United States, the computers and things like that. Nowadays it's not. There is some stuff being made in the US that's

flowing through this investment, but if you actually kind of do the accounting in the GDP numbers, like most of the incremental spend on data centers is actually flowing abroad. So the big beneficiaries of that are Taiwan and Korea to a lesser extent other places, but like that's predominantly where that money is going, which the optimistic view is that if it were to stop, or slow, U.S. production, there would be less of a direct impact than one might naively think if you're not accounting for this imports issue. I don't know if it'll be like that in practice because obviously it would have a big impact on equity market valuations and all things like that. But the idea that the U.S. economy is entirely dependent on AI growth I think is a misreading of what's been happening.

The US economy is chugging along for a variety of reasons. AI is an important impulse, but because so much of it flows straight out the door in higher net imports, it's less clear to me. Obviously any transition might be kind of dramatic and I don't know how you get a transition without it being like some kind of overall downturn, but it's not necessarily as extreme as it was in the 90s. Okay. I mean, link to this is obviously what's going on in the bond market. And I know that's something else you've been writing about. Obviously yields have been rising. The yields are not quite as high as they were in the 90s. I think the 10-year yield got the same level as 2007 or so. So not quite, we're probably higher, I think really. 2023. Was that? It's at the same level, but it's below where it was in 2023 right now.

Okay. Well, sorry. The 30 year yield, I think is... Yeah, 30 years old high. Yeah, 30 years old high. Yeah, you're right. 10 year, I'm ahead of myself. Hasn't broken the 5% level yet. But, you know, obviously, as all of this stuff we've been talking about has, you know, CapEx demand, competition for capital is one of the factors. And I guess strong growth is another factor. Deficits are, you know, is the other thing that people talk about. Obviously, in the late 90s, the US ran a surplus. by 2000. What's your take on what's driving bond yields and are we just part of a normalization back in the 90s kind of a tight period? Have yields further to rise, would you say? I think they could. In general, my sort of simple-minded view is that if you're going to be locking up money for 10, 20, 30 years, getting a fixed nominal return, the nominal return that you would be willing to accept has to be contingent on what you think the alternatives are. So the alternatives, they're basically,

they're a lot, but the two basic kinds of alternatives. One is you hold your money overnight or something very short term. So you're not taking the duration risk, but you're getting your interest right that way. And then, you know, you have some flexibility or you're taking your risky asset, which does not have a guaranteed nominal payout, but has, because it is uncapped upside, but more downside. Those are basically the three choices. There's more than that, but like that's a good kind of very summary, right? So if you think the overall economy is growing at say right now, it's like 7% in dollar terms a year, annualized. Why would you want to lock in a fixed nominal return of 5% for 30 years? I mean, maybe there are reasons why you might, right? The obvious reason is you think it's not gonna stay growing at 7%, right? If you think it's gonna set 7% now and then it's gonna go to 3%, then 5% looks great as a return, right? But That's that's the question. Um, so I don't you know, I don't trade these things myself. Uh, I just have, you know, I just have target day retire and funds, you know, and to try to keep my head out of that too much. But like, I think it's, it's worth content. Like if you think the growth is going to stay around 7% for a while, which it might not, but if you do 5% makes no sense. Um, there are circumstances where it might maybe for a bit, but it seems hard to justify that over a sustained period.

The deficit question I think matters mostly insofar as it flows through to the nominal growth rate. So, at least for a country like the United States. For countries, in other words, in the United States, and for that matter in the UK, Japan, New Zealand, if the government is in danger of running out of money, the central bank can always step in and provide financing if it needs to. that creates the risk potentially of inflation, but it's not going to create the risk of default. There are obviously places where you do a default risk, but that's not, I don't think, the issue in the United States. And so the question is, are deficits creating more growth than they otherwise would maybe? But again, as you pointed out, in the 90s, when interest rates were meaningfully higher than they are now, you had growth that was either the same or faster, and deficits were, there were no deficits, right? There were surpluses.

The way I would think of it is that there's a whole bunch of moving parts in the economy. The government budget deficit is one of them. Households

matter. The business sector, domestic business sector matters. The rest of the world matters. You have to see how those things all add up. What we've seen in the United States over the past several years is that households and businesses have been pretty conservative posts. They've really been conservative since the financial crisis. Since the pandemic, they've been conservative in a different way in the sense that They're spending more than they were in the 2010s, but they're not really borrowing more to finance that and that's the flip side is that the government is borrowing more than otherwise would be larger deficits Foreigners kind of doing the same in the aggregate the same thing they were Different mix of you know, which countries are now but like in the aggregate same so That's why I think you're not seeing the budget deficit itself does not seem to me to be immediately of a concern because I said private indebtedness is falling You could have a situation where private deniness is either stable or rising and the budget deficit is smaller. Not clear to me that would make a difference on interest rates. Not clear to me make a difference like overalls being good or bad thing or why it'd be better. But you could have that as a different mix. But you know, the idea of the budget deficit specifically, I think that the channel is through the extent that it's supporting growth. If you have a bunch of tax hike and spending cuts that hit growth, that hit, you know, the dollar value of growth or real growth and inflation together, that would lower interest rates. But I don't think there's sort of this magical independent thing you can do that would lower the budget deficit, lower interest rates, not hit growth. I think for small countries, there is an option to do that, which is you have the exchange rate depreciate a lot. You look at like the early 1990s and Canada did that, some of the Nordic country like Sweden and Norway and stuff did that after the banking crisis.

Those are smaller countries that can get away with that sort of thing. I don't think the U.S. could for a variety of reasons, but that's like the other mechanism. So I don't want to say it's literally impossible to have that combination, but it's hard and it's not something that you'd expect for a large economy like that. So I mean, it sounds like from your perspective that the rising yields, okay, reflects a number of different factors, but seems justified, okay. Largely by stronger growth, is that fair to say? Yeah, it's not growth and inflation. And also, and the other thing too is, Again, we mentioned briefly, but like the 10-year yield's basically been in a range

bound for the past three years, right? If you look at charts since like mid 2023 to now, it's no trend whatsoever. The third year was a little different, but again, I mean, if you take the sort of optimistic view that we are out of the woods from what happened in the financial crisis, we're back to a kind of faster growth regime. We have some entrenched inflation, which is not great, but it's not like massively terrible. Do you think maybe we're gonna have some kind of productivity boom coming?

It doesn't seem crazy coming to have the interest rates where they are. I mean, people are saying it's the highest interest rates in 19 years. Like, okay, well 19 years is a long time. And money of the intervening years between now and then were actually quite terrible. And that's why rates were low. So, you know, it's good to have some perspective. I think actually this also explains Japan, right? The Japanese tenure, I think at the highest in 30 years, like, wow, it's crazy. Like Japan is famously known for having had lost decades. Trying to get out of a situation where they had essentially zero growth in young terms. no falling prices, seems like they finally managed to make it happen. Good for them. It's not surprising that if that were the case and people believe it's the case, that interest rates would be somewhat different than they were when people thought Japan was just going to shrink forever into zero. So I think it's good to have some perspective on like, take the win, I guess I would say. These things, there are worlds where interest rates go up and it's not bad. I think this is one of those times.

As we've been saying, there was no deficits at the end of 2000, and I think the debt to GDP ratio was maybe, I don't know, 40% of GDP or something like that. So debt service costs is a concern. Obviously, people, economists talk about R versus G, and okay, if growth is going to be 5, 6, 7%, then that's still okay. But how much of a concern is debt sustainability and these level of rates given the debt stock that the US has? These are all good questions. I think, again, to your point, as of now, G is still higher than R. The issue is that the deficit excluding interest is big enough that it doesn't matter. So that's why the debt stock is still growing relative to the economy.

I guess the question is, does it matter that debt is growing relatively? People talk about the snowball effect, like it's something inherently. That's going to be true for any sector of the economy that borrows. And some sector of the

economy is going to be borrowing as long as the United States is running a current account deficit, which is by definition what's going to happen if the rest of the world and the aggregate is going to be running a current account surplus. In other words, if everyone in the rest of the world collectively wants to buy more financial assets than they sell, then somebody has to be selling more financial assets than they buy. And that seems to be the United States, not exclusively the United States, but it has been for a while. There are various reasons for that. If that's going to be the case, surely it is better for the entity selling financial assets to be the government, which is going to have the lowest cost of capital, the least risk of being able to roll anything over, the easiest ability to raise money than the private sector, right? Because that would be the alternative. That's sort of my general view on this. Obviously, if you know, you don't have the U.S. running a current account and deficits a different story, but that would be sort of my overall perspective. In terms of debt service costs, as of now, debt service costs relative to the economy are still lower than at the peak in the early 1990s. That's because a lot of the debt, first of all, interest rates were a lot higher in the 80s and early 90s. And the debt stock, I think in the 80s, peaked at something on the order of 60 to 70% of GDP. It was quite a bit higher than it ended up falling and then interest rates were higher and those interest rates were locked in for a long time. We still have a lot of debt at low rates. Now that had been said, I'm sure debt service will rise relative to where it is now, just given how interest rates are staying where they are and older debts at lower rates are going to get rolled. But I think it's just useful to have that context. How much does that matter overall? Again, I think it depends more on the alternatives. I think an unfortunate situation to be in which I don't think is where we are, but like the Brazil scenario, which is where you have real interest rates that are extraordinarily high. You have essentially balanced budgets excluding interest, but debt service and interest costs are very, very high in part because interest rates are so high. So it's a really bad equilibrium that they find themselves stuck in. And that's a legacy of having several hyperinflation episodes in the past few decades. They haven't had any since then, but like, I think that's a big explanation for that. We're not there. I hopefully won't be there. I mean, that is like a potential situation to be wary of. But I don't think we're anywhere near that right now. Good soulful. Maybe just shifting gears a little bit. And obviously, I know you've written the book, Trade Awards or Class Wars with Michael Pettis. And

we've had Michael on as well. And obviously, big focus of that is global imbalances.

And that is back in the news again I guess we've just come out of the G20 and obviously there was a kind of a general agreement excluding China underneath to address these things. There was also an article in the Wall Street Journal from Greg Ip suggesting that we may need another plaza cord. I mean it doesn't sound like the Chinese would be up for that but I mean what was your read on the G20 all of that? How do you see these imbalances? I don't think we're getting any closer to addressing them or not. Sure. So, again, a lot there. I think in general, it's good to help zoom out as to what imbalances are. What do you mean we talked about that? And basically, at the end of the day, every society produces a bunch of things, goods and services, and people in that society, households and businesses and governments, buy a lot of goods and services. And invariably, those things do not have 100% overlap.

because we don't have any fully autarchic societies and people trade back and forth across borders and that's fine. And sometimes the aggregate amounts don't match either and you cover the differences by borrowing or lending, right? It's not as if people decide as a country, we're going to be usually, right? Like we're going to be net borrowers or net savers or what have you. It's individual entities and they just kind of add up and we draw lines around where people are based on jurisdictions. And then we say whether you have a deficit or a surplus, but that's essentially what's happening here. Some of that's totally normal, right? There's nothing, there's no inherent reason why at any given point in time every economic end of the day should always be exactly balanced in how much they spend and they earn. There are plenty of good reasons why if you think, for example, your income is going to go up a lot if you make a certain kind of purchase that you would borrow to invest or issue equity to invest, right? We're seeing, we're talking about AI catbacks earlier, like very good example, right? If you believe that that's going to make you a ton of money in the future, you sell assets now, financial assets now to buy real assets that are going to improve that. Or conversely, you know, you're retired or getting ready to retire and you want to have a bunch of assets and then you accumulate those assets when you have the chance and then draw those

down when you're older, you know, to pay for your living standards, right? That's again, all very reasonable things. The issue is when for whatever reason you have large enough you know, imbalances, right? You know, either lots of saving or lots of dissaving in ways that are fundamentally destabilizing for the global economy. And so there are lots of ways that can happen. We talk about this in the book, which I definitely encourage people to read if they want more on the mechanics or in the history of this for that matter. But essentially, if you have lots of essentially inequality shifts in the distribution of earning power, then you have to have lots or concentration of earning power, you have spending power.

you have to have lots of debt to balance that out, right? Like in sort of an extreme example, let's say one person earned all the money, but how do they earn the money, right? They have to earn it because someone's buying stuff. But if no one has any money to buy stuff, they have to borrow to buy it, right? Again, this is a very extreme style. That's not how the real world works, but like there are versions of this that are kind of more realistic than not. And so, and then that's a potentially a problem because if people are borrowing a lot, to finance their current spending, that's very different from borrowing to buy assets that are going to create value in the future. You have to be able to keep borrowing to finance your current spending. That borrowing is not going to end up creating more value for you in the future. It's better than not borrowing because you starve, but it creates potential. So the reason people care about imbalances, I think the reason you should care about imbalances, are twofold. One, to what extent do large imbalances like this reflect problems in the distribution of income and people potentially living below their means? And then to what extent do these distributions, maldistributions of income are offset or correspond to big increases in indebtedness that threaten financial stability? And so, We saw really good examples of this in the period leading up to the global financial crisis. We've seen examples of this before. The 1920s actually buried examples well. So what's happening now is, in many ways, sort of a recapitulation of things we saw in the 2000s. In particular, in China, for a variety of reasons, people in China are underpaid relative to the value they create, the workers in China.

On top of which there are lots of features of Chinese society, China's political economy that make it so that people, workers or other consumers just don't have the spending power you would expect relative to their productivity. And the net effect of all this, and I'm happy to talk about in detail if you want, but the net effect of all this is that consumer spending in China relative to the value of what's produced in China is extraordinarily low. basically lower than what you see in any other society that's comparable, ever. The only places you can find are comparable in terms of the share of production that's domestic consumption. Our places, like these are the corporate tax havens where the GDP number is inflated by, you know, activities are not really domestic. Oil sheikdoms where, again, like a lot of economic activity is essentially exported or whatever, right? Or places like the Soviet Union under Stalin.

None of these are benign. It's extremely odd. Now, Chinese living standards have obviously been rising rapidly over time because overall growth has been going up so much, but the share is very low. And for a long time, the share was falling. It kind of bottomed out around 2010 and it stayed pretty low. It kind of took another hit with COVID, but the share is very low. And so historically, the way that's balanced out is that, the Chinese economy, Chinese government has encouraged lots and lots of investment in infrastructure and other businesses real estate to offset the weakness of the domestic consumer to create jobs and incomes for people. And in addition, also, even with all that, there's been a lot of exports relative to imports, right? Because that's the other thing. If you produce a lot more than you consume, it has to go somewhere. So it gets exported. Since the pandemic, what's been really striking is China's pandemic response stood out compared to other countries where basically most of the rest of the world, especially if they had the resources to do so, supported workers and consumers, gave people money like, you can't work, it's okay, you can keep spending. China did the opposite. We don't have exact data on this, but you can get a sense from official Chinese data, the number of people who were migrants from the countryside to cities fell something on the order of 50 to 70 million people during the initial lockdowns in 2020, which means people there lost their jobs. They were not eligible for unemployment because they were not official residents of those cities. They had to move home, become subsistence farmers, tens of millions of people, right? And there were

another set of lockdowns in 2022. And in general, growth has been very, very anemic in China since the domestic demand growth, I should say. And that's been offset by more production for exports the rest of the world. That was supported in part by the fact that people in the rest of the world wanted more physical goods. Chinese businesses were happy to accommodate that. In addition, the Chinese government's response is part of their, to the extent that they did do any sort of stimulus to offset the pandemic, mostly with supporting investment in manufacturing facilities. So you have a lot of new factories and subsidies for building and producing more to export. The result of all this has been that China's trade surplus in the aggregate has increased from something on the order of 100 billion to 200 billion dollars a year on the eve of the pandemic to over a trillion dollars a year now. That's pretty substantial. Relative to the global economy is substantial relative to the particular relative to the economy, global economy, ex-China, that is a big shift. And obviously, that is something that some China's trading partners are finding very difficult to deal with because It's one thing to get more stuff from China if you want it and you're able to sell more things in return. That's how trade is supposed to work. You know, I make something new you didn't have before. You think that's great. You buy it. That gives you more money or gives me more money. So I buy some stuff that you have. That's great. We both benefit. That's the whole point of exchange. If you have a situation where I sell you something and then I buy nothing in return. On the one hand, you get the thing and that's kind of nice.

But on the other hand, like, how'd you pay for that, right? And then how do we keep doing this over and over again? And what happens to the people you have in your country who used to sell things, make a living? So that creates a lot of problems. It happened, as I said, it already happened in a lot of places, particularly the US in the 2000s. It's now happening in Europe. I think people are, I mean, it arguably happened in Europe before, but I don't think people really noticed it to the same extent. It's now happening very acutely because of China's exports of motor vehicles and particular chemicals, a few other things. So Europeans have become very aware of this. Obviously the Chinese don't like the idea that it's anything, they're doing anything wrong. I was actually in Dalian at the end of June this year when Premier Li Chang said China, there's no China shock 2.0, it's China opportunity 2.0.

Which okay, it doesn't change the fact that if you're on the other side of this you're losing out Interestingly, this is a very live debate within China There are plenty of people within China experts and academics and official it's even the official position of the Chinese government that domestic consumer spending should rise That's very important that the Chinese economy needs to be tilted more in this direction. It just hasn't happened It's not become enough of a priority to make a you know dent in this phenomenon And so that's what the G20 the G7 had a whole you know issue about this, you know, there was an expert commission that was a few months ago that came out with their report. They made a statement on this, the G20 made a statement about this. You know, now this is, one of the things people have said as well, you can look at China's trade surplus, overall current account surplus, which includes things like investment income and say the China's currency is substantially undervalued. I think that should be relatively uncontroversial. The question is then, The Plaza Accord of 1985 is deeply unpopular in China as an idea. I understand why Greg wrote that column, but that is a non-starter because the Chinese believe that this was essentially a trick that the U.S. played on Japan that destroyed them and take them out of the running as being another great power. And they think that it worked. I don't think that's actually right at all.

But nevertheless, that's what they think. So the idea that you're using, I mean, Fugich Merz of Germany made this point, I think, a few months around the time the G7, the same thing about the Plaza Accord. I don't, I'm not saying that it's wrong to focus on the currency imbalance. I think that's a useful sort of starting point quantitatively to talk about what is the scope of the imbalance? What's the right response to what China has been doing? China clearly has been intervening its currency in a particular way to make it cheaper than it otherwise would be. But, you know, I don't know if that's, but like the politics, I mean, I'm not a politics expert, but like, from what I read, like I don't think it's gonna go, maybe it's still right to focus on the currency, as I said, as a way to calibrate the response, but good luck getting them to adjust. But I mean, that's sort of the overall context of why people are talking about this now. And I think it's reasonable China's trades, as I said, the contribution to China's growth in the past few years from the increase in net exports has been massive.

For an economy of its size, it is essentially unprecedented and is having a lot of negative consequences for the rest of the world. The cleanest answer is for people in China to spend more on themselves, to increase their own living standards. Everyone would be better off if they do that. That sort of solves the problem. If they're not going to do that, that creates challenges for people in the rest of the world. How you respond is complex, but I mean, you see the Europeans in particular are grappling with this. It's not, you know, it's not going to be easy. I mean, what is the sense on that? Obviously, the US has had tariffs, they've gone to engage in a trade war in Europe, and it seems like it hasn't really, well, hasn't posed tariffs in some sectors, but they're not the same extent. So is that the likely response? Do you think that's the escalation? I mean, I'll also say in the US case, it's very odd because if you were to look at the data on...

global imbalances and impact on the US and so forth, then you look at the actual US policy on trade and imbalances over the past 18 months, it doesn't really correspond very well. It seems like most of the energy the US has expended in terms of tariffs and trade conflict have been with countries that are not contributing to global imbalances or they are not doing anything that would really detrimental to the US economic prosperity. I mean, this latest example of the Canada situation is absurd. China has comparatively been let off the hook, actually. It's very odd. Again, going back to when I was in Dalian, one of the things I was most struck by was someone saying, this was the, I think he was the head of McKinsey's Greater China Office. I mean, this is on the public records. I'm not revealing anything. So you're like, you can watch a video of this. But he said that he thought that the business climate and US-China relations now, or as of a month ago, whatever, two months ago, was better than it had been.

in a very long time and much better than it was two to three years ago. In other words, 2023, 2024. So, which is bizarre given what one might naively have thought based on how campaigns were constructed and how people present themselves. So I think that in that sense, the U.S., it's not really done arguably the most constructive things on this front. I think Europe is probably gonna react differently. If you take sort of, I think a very reasonable estimate that the Chinese Yuan is undervalued on the order of 35%, if the Europeans then take that and say, okay, we're going to apply

tariffs on a bunch of sectors on the order of 35, 40%. I mean, we'll see what happens. I mean, there's obviously the possibility that there will be rerouting. That's what happened when the U.S. put tariffs on China in the first Trump administration, essentially direct imports in the U.S. data. And according to the U.S. data, direct imports from China went down a lot, but then imports from much Southeast Asian countries went up a lot, so the net effect was minimal. I mean, something happened. I mean, I'm sure there was technically some transformation and value added that occurred to make it not purely like customs fraud, but it basically just rerouting. It's entirely possible that would happen as well if the Europeans did that with China, maybe not. That's the danger of using tariffs as opposed to the exchange rate.

It seems like there's going to be something happening. I don't know what. I mean, you probably have a better sense than I would, you know, live where you are. But that's, it seems like it's going to happen. But on the kind of, the plans record on the kind of issue, which I agree, you know, it's... In theory, it needs Chinese agreement, but the area where we have seen some movement is obviously in currency markets and intervention, and the treasury intervened with Japan to buy yen, which is obviously a very rare occurrence. I mean, it was framed initially as helping a friend, but what was your take on what was the rationale for the treasury's yen intervention? First of all, it's hard to know exactly how much they actually intervened.

Besson wrote in very big letters and like had people photographing and buy five to 10 billion worth of yen. My former colleagues at FD Alphaville wrote that as of the data that we can see, assuming they reported everything correctly, they bought only 500 million. Either way, like these are very small amounts relative to the size of the currency markets. I think at the broader point is it's understandable why you'd want I mean, so Japan has something on the order of a trillion dollars worth of foreign currency reserves, directly as foreign currency reserves. The Japanese government as a whole has many, many more dollars worth of foreign currency assets that are held by various public sector entities in addition to the official reserves. There is no reason if they wanted to, and those assets were all purchased by the way by the yen was much more expensive. So if they were to sell those assets for yen, you'd book immediately a massive profit if you wanted.

And arguably, that might be worth doing if you think the exchange rate is too cheap and you think imports are too expensive and people are upset about that. They could do that. Other countries could theoretically do the equivalent thing and say yen's cheap. I don't know, right? They haven't done that. I think my sort of sense from reading the newspaper, I don't have any deep inside knowledge here, but like from what other people have written is that Besson didn't want the... Japanese to sell dollar assets because he was afraid of the impact on US Treasury yields from them doing that. And so he thought that if he did something that prevented that or some alternative, that's also why they mentioned that the Japanese could use the Federal Reserve's, one of their facilities where essentially you can, if you own a Treasury, you can go to the Fed. You lend, you give the Treasury to the Fed, you borrow dollar cash and then you can sell it.

Um, so like Besson talked up how they could do that instead of just selling the treasuries outright. Now the problem with doing that from the Japanese perspective is you should pay interest like why do you, you don't need to do that, right? Um, it's more useful if you, if you really use a short-term thing, if you're doing a long-term currency intervention, it's a different story. Uh, but anyhow, I think it's just that he has this fear of anything making treasury yields go up. Uh, at least that's what the news reports, I mean, I don't know, I'm not inside his head and not talking to them, but that's like what the news reports say. Um, I don't know if that makes a lot of sense, but.

But does the US want generally a stronger yen or are they worried about the yen at all? I don't know enough about that specifically. I mean, the Japanese seem like they want a somewhat stronger yen. I mean, they've intervened repeatedly over the past few years. And then the question is, how do you get that? I mean, I'm not sure. So one of the challenges is Japanese yields have gone up, right? And they've gone up relative to other... countries yields. And so naively, you would think that the currency should have appreciated during that time. But that occurred at the same time that inflation and growth expectations in Japan shifted. So if you think about sort of a forward-looking view, maybe the actual change in yields was not as significant as it looked. And so I think that's part of, might be why the yen has not appreciated over the past few years in nominal terms. It was like 160.

150 like four years ago when they first in the Japanese did their first big intervention and back in like September October 2022 Right, and it hasn't really, you know, it's gone up a little bit got down a little bit but hasn't moved a lot And so one might think like that's odd because interest rates in Japan come up a lot particularly the longer end They've gone up quite substantially But as I said if you and I said on a relative to like the dollar and Euro rates and stuff and yet relative to inflation expectations Maybe it hasn't been as much of a move It's hard to say that for sure, but that might explain partly. But like, again, if they want the yen to appreciate for essentially they don't want somewhat because it's hitting their inflation in ways that they find undesirable, they don't want their yen to appreciate so much that it hurts their exports. The domestic conditions in Japan don't justify radical increases in interest rates. I mean, arguably that's what foreign exchange reserves are four, right? That's the combination. Like your domestic conditions do not justify raising interest rates a lot. There's no immediate pressure on interest rates just because of the yen. So you could use the reserves to smooth things out. I get why they might want to do that. I also get why they might not want to do that. You could just do nothing. But to the extent that there's some unpopularity there. But it's a tough question. I mean, overall inflation in Japan is not excessive, especially relative to what they've been targeting.

You know, maybe it gets worse over time, but it's actually they've actually done a pretty decent job We're depending on your point of view, but like it hasn't really it's basically kind of where they want it real growth. Yeah, real wage growth has not been fantastic Not on the wage growth is basically followed inflation, you know in parallel Will the strong currency help? I don't know like I mean one Too much in here, but like one interesting phenomenon of the past five six years is that Japanese companies non-financial companies, because they've accumulated such a massive portfolio of interest-bearing assets, including outside Japan, their overall profits have been massively inflated relative to their operating profits in the past six years as interest rates in the rest of the world have gone up relative to Japanese interest rates. And that's also that. And then I guess maybe the depreciation has been...

Well, there hasn't been a lot of depreciation in the past four years. I've been pretty stable. So the extent that Japan raises interest rates a lot, you would have an impact there. It would hit those kind of, I don't know how much of a difference that really makes in the grand scheme of things, but like that's like a potential factor for them to consider. As again, domestic conditions don't really warrant a dramatic tightening. So, I don't know, hard to say, but I see there's a lot of moving parts there. Why the US would intervene? I mean, yeah, it's better if you think the dollars overvalued in general More expensive like that helps but like there are lots of currency. I mean Korean won is even more obvious example Then yeah, I think so but they haven't done that yet as far as I know OK, just before we wrap up, I just want to come back to what you said at the outset. Obviously, we've got a lot of the task forces that have been appointed by Warsh and you mentioned the Productive View one in particular. I mean, when you look across the different task forces, the personnel in there, do you expect something dramatic to come out of this whole process or not or any thoughts on how this might play out? A lot of people have made this point. The balance sheet task force I think is very interesting potentially.

just because of the caliber of people they have there and the nature of the question and the fact that it's really unsettled. I think, you know, how that's gonna play out. And that's Raghuram Rajan, is it? Yeah, Raghuram Rajan, Jeremy Stein, and Ambulad Karendine. And I think that'll be interesting to see how that... Part of the reason that's interesting is because I think of all the questions that the task forces have been set up to address is the one where there's the most ambiguity among what people currently at the Fed actually think should happen. So, I mean, some of the questions, it's like, I don't know if people at the Fed really, there's a lot of disagreement there or there's like an unanswered question, but it's the balance sheet one. It seems like if you look at what Fed officials have been saying publicly in the past few years, what to do about the balance sheet, how to think about it, what are the options? It just seems like much more of an open-ended question and lots of possible different answers. And so that could go interesting places.

I mean, they could be other interesting questions, other things too, but I would be like, in terms of the ability, like the task force is not binding,

right? At the end of the day, things are still voted on internally by the Fed, but like, if people are just genuinely don't have a strong view or are divided deeply, then maybe that would have kind of more of an impact. That's kind of more speculative on my part, but I think that's particularly one to watch. Let's leave it there. Great to have you back. And for people who want to follow your work, obviously you're right. The overshoot on Substack so people can follow you there and subscribe to get your latest views. But we appreciate you coming back to share your perspective. So from all of us here at Top Traders Unplugged, thanks for tuning in. We'll be back soon with more content and we'll speak to you then. Thank you very much. Thanks for listening to Top Traders Unplugged.

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